Problem Set 6

Due Saturday, October 24, 11:59 PM.

You may use Mathematica or other computational tools in all problem sets.

1. (a) Consider two Hermitian operators $\hat{A}$ and $\hat{B}$ with simultaneous eigenvectors such that $\hat{A}|ab\rangle = a|ab\rangle$ and $\hat{B}|ab\rangle = b|ab\rangle$.

   i. Show that $[\hat{A}, \hat{B}] = 0$. (5 points)

   ii. Show that $[\hat{H}, \hat{P}] = 0$ for a particle in a symmetric potential, where $\hat{H}$ is the Hamiltonian operator and $\hat{P}$ is the parity operator. Considering $[\hat{H}, \hat{P}] = 0$ and $\hat{P}^2 = 1$, show that the wavefunctions in position representation could be chosen to be either even or odd? (5 points)

   (b) Suppose $[\hat{A}, \hat{B}] = 0$ and the eigenvalue a of $\hat{A}$ is 2-fold degenerate, with two orthonormal states corresponding to the same eigenvalue, $\hat{A}|ab^{(1)}\rangle = a|ab^{(1)}\rangle$ and $\hat{A}|ab^{(2)}\rangle = a|ab^{(2)}\rangle$.

   i. Suppose we write $\hat{B}|ab^{(1)}\rangle = b_{11}|ab^{(1)}\rangle + b_{12}|ab^{(2)}\rangle$ and $\hat{B}|ab^{(2)}\rangle = b_{21}|ab^{(1)}\rangle + b_{22}|ab^{(2)}\rangle$, where b_{ij} are constants. Construct two orthonormal states $|ab_1\rangle$ and $|ab_2\rangle$ that are linear combinations of $|ab^{(1)}\rangle$ and $|ab^{(2)}\rangle$ and simultaneous eigenstates of $\hat{A}$ and $\hat{B}$ with non-degenerate distinct eigenvalues b_1 and b_2 for $\hat{B}$, and determine b_1 and b_2 in terms of b_{ij}. (5 points)

   ii. Now suppose b_{ij} are such that after doing (1(b)i) it turns out $b_1 = b_2 = b$ and the states are still degenerate, $|ab_1\rangle = |ab_2\rangle = |ab\rangle$. Can you distinguish the two states, and what would you need to construct states that are not completely degenerate? (5 points)

2. Consider a physical observable which can take only two possible values and represented by operator $\hat{A}$, and a second physical observable which can, again, take only two possible values and represented by operator $\hat{B}$. The eigenstates $|a_1\rangle$ and $|a_2\rangle$ of $\hat{A}$ and $|b_1\rangle$ and $|b_2\rangle$ of $\hat{B}$ are related by $|a_1\rangle = \frac{3}{5}|b_1\rangle + \frac{4}{5}|b_2\rangle$ and $|a_2\rangle = \frac{4}{5}|b_1\rangle - \frac{3}{5}|b_2\rangle$.

   (a) A measurement of $\hat{A}$ (i.e., the observable represented by operator $\hat{A}$) produces a_1. What is the state of the system immediately after the measurement? (5 points)

   (b) Right after the measurement of $\hat{A}$ in part (2a), a following measurement of $\hat{B}$ is done. What are the possible outcomes and with what probability? (5 points)

   (c) Right after the measurement of $\hat{B}$ in part (2b), a following measurement of $\hat{A}$ is done. What is the probability for obtaining a_1? (5 points)

(d) Question (2c) was answered without knowing the outcome in the measurement of $\hat{B}$ in part (2b). Suppose b_2 was measured in part (2b). What would have been the probability for measuring a_1 in part (2c)? (5 points)

3. Using the raising and lowering operators, answer the following questions for a particle of mass m in the one-dimensional harmonic oscillator with potential $V(x) = \frac{1}{2}m\omega^2 x^2$.

   (a) The uncertainty in position, Δx, in stationary state $|n\rangle$. (5 points)

   (b) The uncertainty in momentum, Δp, in stationary state $|n\rangle$. (5 points)

   (c) Check whether the product $\Delta x\,\Delta p$ obeys the uncertainty principle. (5 points)

   (d) Rewrite the Hamiltonian $\hat{H}$ in terms of $\hat{\alpha}_R$ and $\hat{\alpha}_I$ defined by $\hat{x} = \sqrt{\frac{2\hbar}{m\omega}}\,\hat{\alpha}_R$ and $\hat{p} = \sqrt{2m\hbar\omega}\,\hat{\alpha}_I$. Rewrite the uncertainty relation in stationary state $|n\rangle$ obtained in part (3c) for $\Delta\alpha_R\Delta\alpha_I$. (5 points)

   (e) When the oscillator is not in a stationary state, or an eigenstate of the Hamiltonian operator, the uncertainty would take other values than those for individual stationary states obtained above. Show the boundary that separates the region that satisfies the uncertainty principle and the region that does not on a plot of $\Delta\alpha_I$ versus $\Delta\alpha_R$. Indicate the points $(\Delta\alpha_R,\ \Delta\alpha_I)$ on the plane for the first four stationary states corresponding to $n = 0, 1, 2$, and 3. (5 points)

4. A coherent state (or Glauber state) $|\alpha\rangle$ of the simple harmonic oscillator is an eigenstate of the lowering operator, $\hat{a}|\alpha\rangle = \alpha|\alpha\rangle$.

   (a) Determine the coherent state as a superposition of the stationary states $|n\rangle$ and normalize it. (5 points)

   (b) What is the probability that a system in a coherent state would be found in stationary state $|n\rangle$? What kind of probability distribution function of n is it? (5 points)

   (c) Determine the uncertainties Δx, $\Delta\alpha_R$, Δp, and $\Delta\alpha_I$, using the raising and lowering operators, in the coherent state. (5 points)

   (d) Show the point for uncertainties $(\Delta\alpha_R,\ \Delta\alpha_I)$ for the coherent state on the plot you made in part (3e). Comparing $\Delta\alpha_R$, $\Delta\alpha_I$, and their product, what is special about the coherent state? (5 points)

5. How long did it take you to complete this problem set?